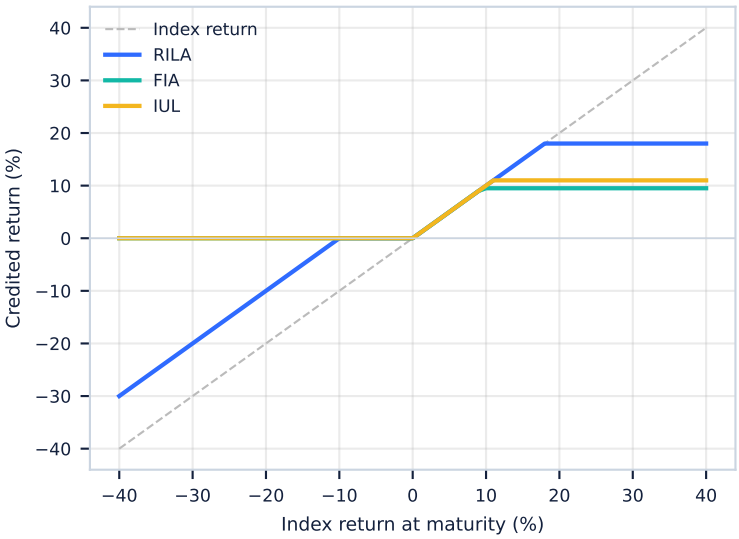
Simple ALM hedging model: RILA, FIA, and IUL embedded options

Illustrative model with synthetic inputs. One year, one index, closed-form Black-Scholes, $100 of account value per product.
Synthetic market: index 100, 1y rate 4.0%, dividend yield 1.5%, implied vol 19%, option budget $3.50 per $100.

Payoff at one year per $100 of account value



How to read the products

RILA: 10% buffer, 18.0% cap: long ATM call, short 1.18 call, short 0.90 put. FIA: 0% floor, 9.5% cap: long ATM call, short 1.095 call. IUL: 0% floor, 11.0% cap on one year of account value: long ATM call, short 1.11 call. The policyholder is long these packages, so the insurer carries them as liabilities and must fund them from the general-account spread (the option budget). The RILA's short buffer put is what lets its cap sit well above the FIA and IUL caps for a similar cost.

Pricing and Greeks (per $100 account value)

	RILA	FIA	IUL	Portfolio
PV	3.4102	3.8263	4.2827	11.5192
Delta (per index pt)	0.5392	0.1859	0.2129	0.9381
Gamma (per index pt^2)	-0.0119	0.0001	0.0005	-0.0112
Vega (per vol pt)	-0.2259	0.0022	0.0100	-0.2137
Rho (per bp)	0.0051	0.0015	0.0017	0.0082
Theta (per day)	0.0026	-0.0009	-0.0012	0.0004
Cap	18.0%	9.5%	11.0%	
Cap for $3.50 budget	18.47%	8.50%	8.50%	

Full-revaluation stress P&L, insurer view ($, negative = loss)

Scenario	Liability PV	Unhedged	Futures	Hedged
Index -20%	-8.10	+19.62	-18.76	+0.86
Index -10%	1.77	+9.75	-9.38	+0.37
Index +10%	20.16	-8.64	+9.38	+0.74
Index +20%	26.84	-15.32	+18.76	+3.44
Vol +5 pts	10.24	+1.28	+0.00	+1.28
Rates +50 bp	11.93	-0.41	+0.47	+0.06
Rates -50 bp	11.11	+0.41	-0.47	-0.05
Range (max - min)		**34.94**		**3.50**

Hedge recommendation and residual risks

Buy 0.915 index futures units (notional about $91 against $300 of account value) to offset the liability delta of 0.938 per index point. Full revaluation shows the stress P&L range shrinking from $34.94 to $3.50. The hedge is linear, so residual risks remain: the insurer is long gamma (+0.0112 per index point squared, so the hedged book gains on large moves in either direction; worst hedged scenario is Rates -50 bp at $-0.05), long vega (+0.214 per vol point; a +5 pt vol shock adds $1.28), and rho of -0.0082 per basis point, which the futures roll partly offsets. Rebalance the futures as delta drifts with the index and with time, and revisit the hedge if the gamma, vega, or rate residuals move outside the risk budget. Illustrative model with synthetic inputs.